

Tejas Kulkarni

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Personal Website || Github || LinkedIn
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Mechanical Engineering Undergrad

EDUCATION

BITS PILANI, GOA CAMPUS | B.E. (HONS.) MECHANICAL ENGINEERING

2024 – Present | Goa, India

Current CGPA: 9.20 / 10.0

Relevant Coursework: BITS F441 Robotics, CS F320 Foundations of Data Science

RESEARCH & PROJECTS

OPTICS & BIOPHOTONICS LAB, BITS GOA | UNDERGRADUATE RESEARCHER

Aug 2025 – Present

Working on photon statistics and signal processing for delayed luminescence in biological systems, supervised by Dr. Yuriy Petenev. Focus on noise reduction, curve fitting, and interpretation of low-intensity optical signals. Manuscript in preparation.

ADDITIVE MANUFACTURING FOR OPTICS | DESIGN & FABRICATION PROJECT

Nov 2025

Designed and 3D-printed optomechanical components for biological experiments where commercial parts were cost-prohibitive or unavailable. Addressed tolerance, deformation, and repeatability constraints.

LOW-COST WIND TUNNEL | INDEPENDENT ENGINEERING PROJECT

Nov 2025 – Dec 2025

Built a functional miniature wind tunnel using salvaged materials under a \$2 budget. Achieved stable laminar flow and smoke visualization for aerodynamic demonstrations and experimentation.

ACADEMIC EXPERIENCE

BITS PILANI, GOA CAMPUS | TEACHING ASSISTANT — ENGINEERING DESIGN

Aug 2025 – Dec 2025

Conducted lab sessions for first-year students covering CAD, prototyping, and design iteration. Guided students through SolidWorks-based modeling and fabrication workflows. Awarded Certificate of Merit.

ACHIEVEMENTS

ICIST 2025 | BEST SPEAKER AWARD

Karnataka, India

Awarded for presentation on delayed luminescence in plants, evaluated on technical depth and clarity of delivery.

SKILLS

TECHNICAL

Engineering & Computing:

Python • MATLAB • SolidWorks • AutoCAD • 3D Printing • Signal Processing • Numerical Analysis

OTHER

Strengths:

Technical Writing
Research Presentation
Lab Instruction

ADDITIONAL EXPERIENCE & INFORMATION

TEAM GIGAWATT — BITS PILANI EV CLUB | MECHANICAL SUBSYSTEM MEMBER

2024 – Present

Worked on conceptual design, mechanical modelling, and basic simulation tasks for institute-level electric vehicle projects. Exposure to real-world design constraints and team-based engineering workflows.

ABHIGYAAN – SOCIAL SERVICE CLUB | VOLUNTEER

2024 – Present

Conducted regular English teaching sessions for campus housekeeping staff. Focus on clear communication and structured instruction.

FILM MAKING & PHOTOGRAPHY CLUBS | CINEMATOGRAPHY & EVENT DOCUMENTATION

2024 – Present

Contributed to short films and institute event coverage. Experience with visual storytelling, shooting, and basic editing.

COMPETITIVE EXAMS | JEE ADVANCED 2024

India

Secured All India Rank 11,864 out of approximately 200,000 candidates (Top 6%).

LANGUAGES | SPOKEN & WRITTEN

English • Hindi • Marathi

German (A1)